PROGRAMMING FUNDAMENTALS

C++ Programming Assignment

Assignment No. 2

Department of Computer Science

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Solved C++ Programs

Each solution uses the basic **if** statement and follows the conditions given in the assignment.

1. Student Scholarship

Enter the student's marks and attendance percentage. Scholarship eligibility requires marks of at least 85 AND attendance of at least 90%.

```
#include <iostream>
using namespace std;

int main()
{
    int marks;
    float attendance;

    cout << "Enter marks: ";
    cin >> marks;

    cout << "Enter attendance percentage: ";
    cin >> attendance;

    if (marks >= 85 && attendance >= 90)
    {
        cout << "Eligible for Scholarship";
    }

    return 0;
}
```

2. Online Shopping Free Delivery

Free delivery is offered when the shopping amount is Rs. 3,000 or more.

```
#include <iostream>
using namespace std;

int main()
{
    float amount;

    cout << "Enter total shopping amount: Rs. ";
    cin >> amount;

    if (amount >= 3000)
    {
        cout << "Congratulations! You Qualify for Free Delivery";
    }

    return 0;
}
```

3. Mobile Battery Alert

Show the low-battery message when the battery percentage is 15 or less.

```
#include <iostream>
using namespace std;

int main()
{
    int battery;

    cout << "Enter battery percentage: ";
    cin >> battery;

    if (battery <= 15)
    {
        cout << "Low Battery! Please Charge Your Phone";
    }

    return 0;
}
```

4. Library Book Fine

A late-return message is displayed when the number of overdue days is greater than zero.

```
#include <iostream>
using namespace std;

int main()
{
    int overdueDays;

    cout << "Enter number of overdue days: ";
    cin >> overdueDays;

    if (overdueDays > 0)
    {
        cout << "Late Return: Fine Applicable";
    }

    return 0;
}
```

5. Hospital Emergency Alert

Display the alert when the patient's temperature is 39 degrees Celsius or higher.

```
#include <iostream>
using namespace std;

int main()
{
    float temperature;

    cout << "Enter temperature in Celsius: ";
    cin >> temperature;

    if (temperature >= 39)
    {
        cout << "High Fever Alert: Seek Medical Assessment";
    }

    return 0;
}
```

6. Parking Fee Discount

The discount applies only when the driver has a valid student ID (1 = yes, 0 = no) AND parking time is at least 2 hours.

```
#include <iostream>
using namespace std;

int main()
{
    int studentID;
    float hours;

    cout << "Valid student ID? (1 for yes, 0 for no): ";
    cin >> studentID;

    cout << "Enter parking hours: ";
    cin >> hours;

    if (studentID == 1 && hours >= 2)
    {
        cout << "Student Parking Discount Applicable";
    }

    return 0;
}
```

7. Office Attendance Warning

Display the warning if attendance is below 75%.

```
#include <iostream>
using namespace std;

int main()
{
    float attendance;

    cout << "Enter attendance percentage: ";
    cin >> attendance;

    if (attendance < 75)
    {
        cout << "Attendance Warning";
    }

    return 0;
}
```

8. Smart Home Temperature Control

Turn on the cooling system when room temperature exceeds 30 degrees Celsius.

```
#include <iostream>
using namespace std;

int main()
{
    float temperature;

    cout << "Enter room temperature: ";
    cin >> temperature;

    if (temperature > 30)
    {
        cout << "Turn On the Cooling System";
    }

    return 0;
}
```

9. Bank Minimum Balance Alert

Display the alert when the current balance is below Rs. 5,000.

```
#include <iostream>
using namespace std;

int main()
{
    float balance;

    cout << "Enter current bank balance: Rs. ";
    cin >> balance;

    if (balance < 5000)
    {
        cout << "Minimum Balance Alert";
    }

    return 0;
}
```

10. Road Speed Monitoring

Display a violation message when vehicle speed is greater than the speed limit.

```
#include <iostream>
using namespace std;

int main()
{
    float speed, speedLimit;

    cout << "Enter vehicle speed: ";
    cin >> speed;

    cout << "Enter speed limit: ";
    cin >> speedLimit;

    if (speed > speedLimit)
    {
        cout << "Speed Limit Violated";
    }

    return 0;
}
```

11. Online Exam Time Alert

Display the warning when 5 minutes or less remain in the exam.

```
#include <iostream>
using namespace std;

int main()
{
    int remainingTime;

    cout << "Enter remaining exam time in minutes: ";
    cin >> remainingTime;

    if (remainingTime <= 5)
    {
        cout << "Warning: Exam Is About to End";
    }

    return 0;
}
```